John S. McAlister

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Education

May 2026 **Ph.D. in Mathematics** — University of Tennessee - Knoxville
Mathematical Biology concentration

May 2023 **M.S. in Mathematics** — University of Tennessee - Knoxville
Concurrent with PhD program

May 2021 **B.S. in Zoology** — The Ohio State University
With Honors Research Distinction

May 2021 **B.S. in Mathematics** — The Ohio State University
Mathematical Biology concentration

Research Interests

I study mathematical biology and game theoretic modeling. Using tools from PDEs, nonlocal equations and graph theory my main focus is the interaction between relational structure and game theoretic dynamics.

Papers

In review **McAlister J.S.**, Burns N. (2026) *Myopic Best Response Replicates the Mean Curvature Flow*. Preprint `https://doi.org/10.48550/arXiv.2609.37803`

In review **McAlister J.S.** (2026)*Structured Coordination on Planar Graphs: a Dual Approach*. Preprint `https://doi.org/10.48550/arXiv.2607.22865`

In review **McAlister J.S.**, Fefferman, N.H. Mengesha, T.A. (2025) *Nonlinear Nonlocal Diffusion Equation for the Analysis of Continuous Coordination and Anti-coordination Type Games*. Preprint. `https://doi.org/10.48550/arXiv.2506.13929`

November 2026 Wright J., **McAlister J.S.**, Almeida R., Bletz M., Gray M.J., Lockwood J.L., Masecar S., Piovia-Scott J., Warwick A., Fefferman N.H. (2026) *Trade network surveillance strategies*. Ecosphere. `https://doi.org/10.1002/ecs2.70787`

October 2026 Dadlani R. **McAlister J.S.**, Schwarze A., Kawikatsu M., Iams S. Eissa T.L., Fefferman N.H.(2026) *Planned Behavior, Perceptual Biases, and the Dynamics of Collective Action*. Physical Review E `https://doi.org/10.1103/l3vx-x5yf`
— Co-first author with R. Dadlani as a mentor for an undergraduate research project

December 2025 **McAlister J. S.**, Brunner, J.L., Galvin, D. J., Fefferman, N.H. (2025) *A Game Theoretic Treatment of Contagion in Trade Networks*. PLOS computational Biology. `https://doi.org/10.1371/journal.pcbi.1013845`

April 2025 Kirkland, S., Li, C., **McAlister, J.S.**, and Zhang, X. (2025) *Edge Addition and the Change in Kemeny's Constant*. Discrete Applied Mathematics. `https://doi.org/10.1016/j.dam.2025.04.031`

January 2025 **McAlister, J.S.**, Fefferman N. H. (2025) *Insights into the coordination game with neutral options through simulation*. Dynamic Games and Applications`https://doi.org/10.1007/s13235-024-00612-4`

January 2025 **McAlister, J.S.**, M.J. Blum, Y. Bromberg, N.H. Fefferman, Q. He, E. Lofgren, D.L. Miller, C. Schreiner, K. Selcuk Candan, H. Szabo-Rogers, and J. M. Reed (2025) *An Interdisciplinary Perspective of the Built-Environment Microbiome.* FEMS Microbiology Ecology `https://doi.org/10.1093/femsec/fiae166`

December 2023 Fefferman, N.H., **McAlister, J.S.,** Akpa, B.S., Akwataghibe, K., Azad F.T., Barkley K., Bleichrodt, A., Blum M.J., Bourouiba, L., Bromberg, Y., Candan K.S., Chowell, G.,Clancey, E., Cathroan, F.A., DeWitte, S.N., Fernandez, P., Finnoff, D., Flaherty, D.T.,Gibson, N.L., Harris, N., He, Q., Lofgren, E.T., Miller, D.L., Moody, J., Muccio, K.,Nunn, C.L., Papeş, M., Pachalidis, I.Ch., Pasquale, D.K., Reed, M.J., Rogers, M. B.,Schreiner, C. L., Strand E.B., Swanson C.S., Szabo-Rodgers, H. L., and Ryan, S. J. (2023) *A New Paradigm for Pandemic Preparedness.* Current Epidemiological Reports. `https://doi.org/10.1007/s40471-023-00336-w`

April 2022 **McAlister, J.S.**, Hamilton, I. (2022) *An Adaptive Dynamic Model for the Vigilance Game in Group Foragers.* Journal of Theoretical Biology. 538:111033. `https://doi.org/10.1016/j.jtbi.2022.111030`

Invited Talks

April 2026 **McAlister J. S.**(2026) *Non-local models in Eco-Evolutionary Processes* Interfaceing Ecology and Evolution. Mini Symposium at Princeton University, Princeton, New Jersey

February 2026 **McAlister J. S.** (2026)*On Structured Models of Coordinating Systems* Invited talk for the Arizona State University School of Math and Natural Sciences, Phoenix Arizona

August 2025 **McAlister J. S.** (2026)*Replicator Dynamics with Spatial Structure* Invited talk for the NSF APPEX center summer Research Symposium, Knoxville Tennessee

Posters and Presentations

February 2026 **McAlister J. S.**(2026)*Structured Coordination Games in the Plane* Talk given in the Geometric Analysis Seminar at the University of Tennessee

January 2026 **McAlister. J. S.**(2026)*Replicator Dynamics for Games with Relational Structure* Talk given in the AMS Contributed Paper Session on Operations Research, Game Theory, and Social Sciences at The Joint Math Meetings 2026 in Washington DC

December 2025 **McAlister. J. S.**(2025)*Replicator Dynamics for Games on Networks* Talk given at The Conference for Complex Networks and Applications 2025 at SUNY Binghamton

September 2024 **McAlister. J. S.**(2024)*Structured Coordination in Continuous Spatial and Strategic Domains* Talk given at AMETHYST: Game Theory in Complex Systems during the Conference on Complex Systems 2024 at the University of Exeter

March 2024 **McAlister J. S.**(2024)*The Structured Coordination Game with Neutral Options* Talk given at The Mathematical Association of America - South East Section Meeting at the University of Tennessee - Knoxville

April 2023 **McAlister J. S.**(2023) *An Adaptive Dynamic Model for a Vigilance Game among Group Foragers.* Talk given at the SIAM Graduate Research Showcase at the University of Tennessee-Knoxville

Funding

Proposals

April 2024 **New Techniques for the Analysis of Coordination in General Discrete and Continuous Domains** Submitted to DARPA DSO Critical Orientation of Mathematics to Produce Advancements in Science and Security (COMPASS) (DARPA-EA-25-02-03) **Selected to give Oral Proposal Package** (OPP)

Support

Spring 2025 GRA supported by NSF(DBI) #2312115 **PIPP Phase II: Analysis and Prediction of Pandemic Expansion (APPEX)**

Fall 2024 GRA supported by NSF(DEB) #2207922 **Socioeconomic and Epidemiological Drivers of Pathogen Dynamics in Wildlife Trade Networks**

Fall 2024 GRA supported by NSF(DRL) #2247074 **Developing and Early Understanding of Contagion in Preschool- and Kindergarten-Aged Children**

Fall 2023 GRA supported by NSF(CCF) #2200140 **PIPP Phase I: Predicting Emergence in Multidisciplinary Pandemic Tipping-points(PREEMPT)**

Awards

November 2026 **Blavatnik Regional Award for Young Investigators** - nominee
— Nominated by Princeton University for excellence in postdoctoral research
April 2026 **Spike D. Tickle STEM Endowed Fellowship** - Winner
— Awarded for excellent research by a graduate student in a STEM field
February 2024 **Eaves Teaching Award** - nominee
— Nominated for excellence in teaching among late career graduate students
April 2023 **Eaves Teaching Award** - Finalist
— Awarded for excellence in teaching among early career graduate students
April 2023 **Math GTA Teaching Excellence Fellowship** - Winner
— Nominated for commitment to further the teaching mission of the University.
August 2021 **Academic Performance Assistantship** - Winner
— Awarded for meeting academic milestones in the PhD program early.

Workshops

August 2026 **Evolutionary Games: Mathematical Theory and Biological Insights** National Institute for Theory and Mathematics in Biology (NITMB): Chicago, IL.
May 2022 **CBMS conference: Interface of Mathematical Biology and Linear Algebra** University of Central Florida, Orlando, FL.

Research Experience

Aug 2026 - Present	**Postdoctoral Research Associate — Levin Lab** Princeton University, Princeton, NJ. Advisor: Prof. Simon Levin
Aug 2024 - Present	**Associate Researcher — NSF Center for Analysis and Prediction of Pandemic Expansion** Arizona State University, Phoenix, AZ.
Jan 2022 - Aug 2026	**Graduate Research Associate — Fefferman Lab** University of Tennessee - Knoxville, Knoxville, TN. Advisor: Prof. Nina Fefferman
May 2018 - May 2021	**Undergraduate Researcher — Hamilton Lab** The Ohio State University, Columbus, OH. Advisor: Prof. Ian Hamilton

Mentorship and Teaching Experience

Oct. 2026 - Present	**Math Undergraduate Research Mentor** Princeton University, Princeton NJ.
July 2024 - Aug. 2026	**Applied Math Undergraduate Research Mentor** University of Tennessee - Knoxville, Knoxville, TN.
Aug 2022 - Dec. 2023	**Graduate Teaching Associate - Instructor of Record** MATH 113 University of Tennessee-Knoxville, Knoxville, TN.
Aug 2021 - May 2022	**Graduate Teaching Assistant** MATH 119, 125 University of Tennessee-Knoxville, Knoxville, TN.
Aug 2019- May 2021	**Undergraduate Teaching Assistant** MATH 1075, 1149, 1150 The Ohio State University, Columbus, OH.

Leadership and Volunteerism

2022-2024	**Member**—Graduate Teaching Assistantship Advisory Council
2023-2024	**Senator**—Graduate Student Senate
2022-2023	**President**—Math Graduate Student Council

Software Experience

Proficient

- R
- Python
- MatLab
- LaTeX
- Mathematica

Experienced

- C#
- Java
- Maple
- JMP
- NetLogo